



Geometrically Accurate Feed-Forward Gaussian Splatting for Unbounded Scenes

Master Thesis Defense

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Prof. Dr. Gerard Pons-Moll

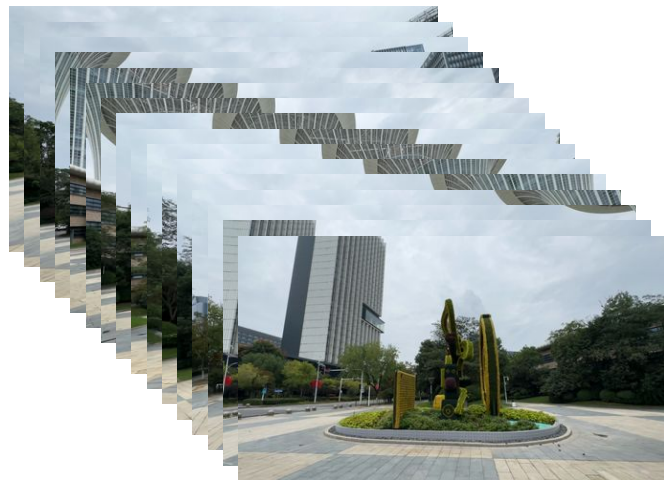
Real Virtual Humans, University of Tübingen

1

Introduction

Feed-Forward Gaussian Splatting

Feed-Forward Gaussian Splatting models take in multi-view images and output a Gaussian Splatting representation

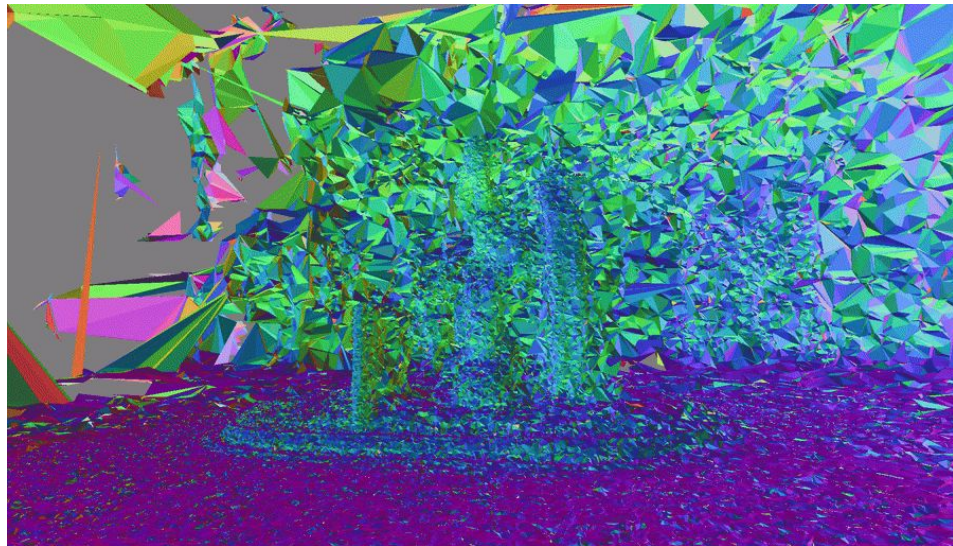


$\sim 0.4s$

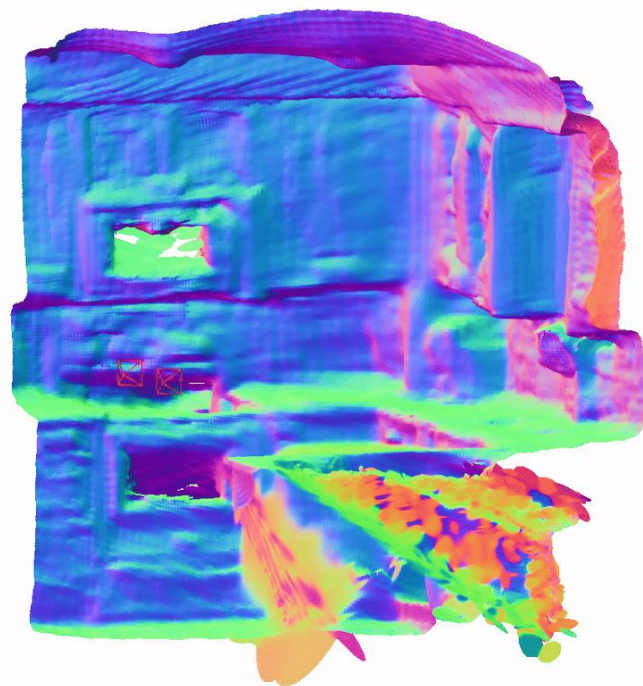


Motivation

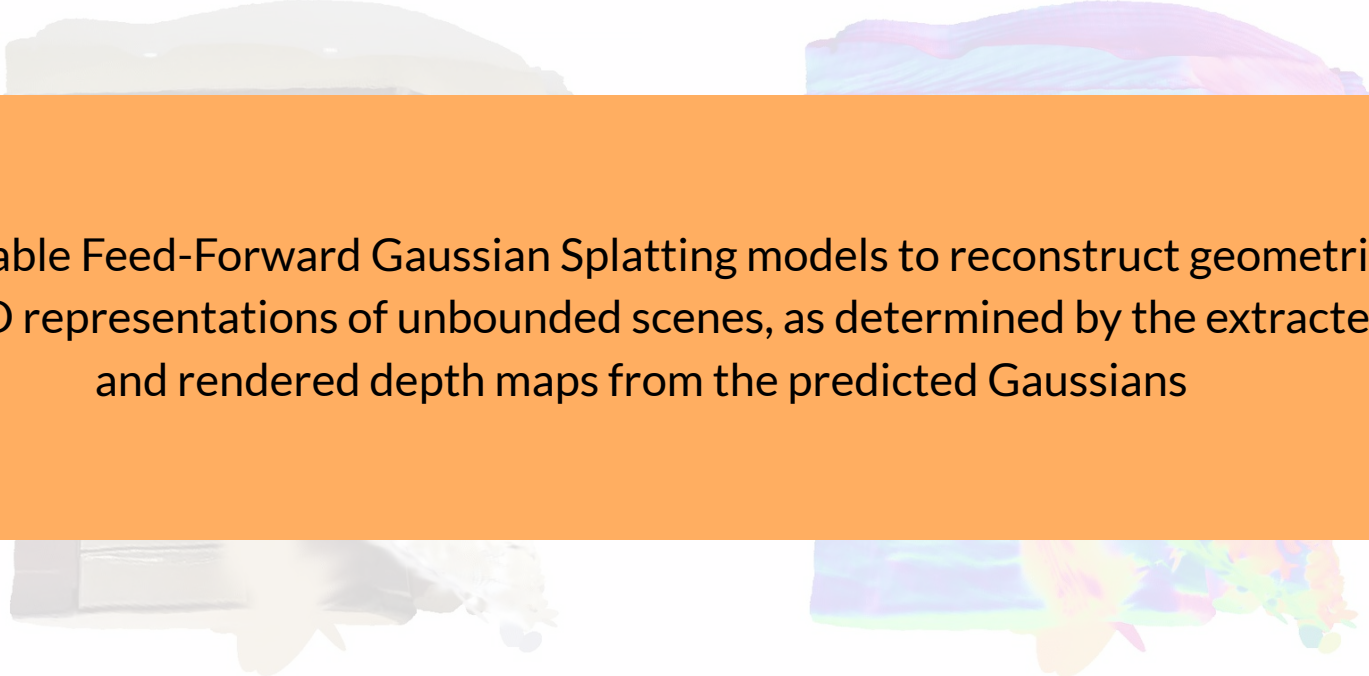
Feed-Forward Gaussian Splatting models achieve great visual quality but fail to achieve geometric quality, especially in the unbounded scene case



Motivation



Motivation



Goal: Enable Feed-Forward Gaussian Splatting models to reconstruct geometrically accurate 3D representations of unbounded scenes, as determined by the extracted mesh and rendered depth maps from the predicted Gaussians

Index

1. Introduction

- **Data:** Unbounded scenes
- **Model**

2. Related Work

- Gaussian Splatting Renderers

3. Method

- Our additions to enable geometrically accurate 3D representations

4. Evaluation

- Metrics
- Extracted mesh and rendered depth maps
- Findings

Datasets: Training

DL3DV

- 10,510 multi-view scenes
- Around half of the scenes are unbounded
- Does not provide complete GT depth

Depth Anything 3 massive dataset

- 160k environmental scenes
- 5% are outdoor
- 20% come from mixed datasets: indoor or outdoor
- GT depth comes from SfM methods thus incomplete

Datasets: Training

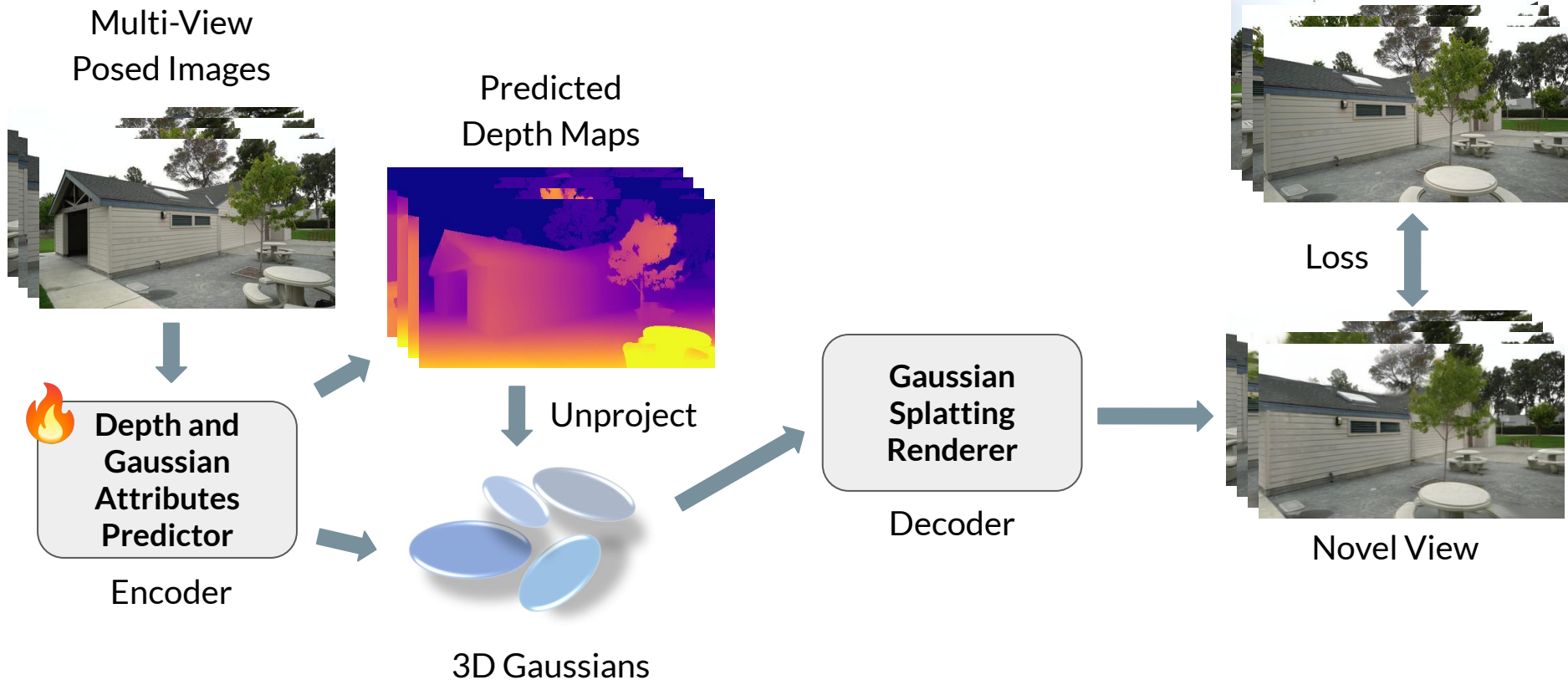
DL3DV

- 10 510 multi-view scenes

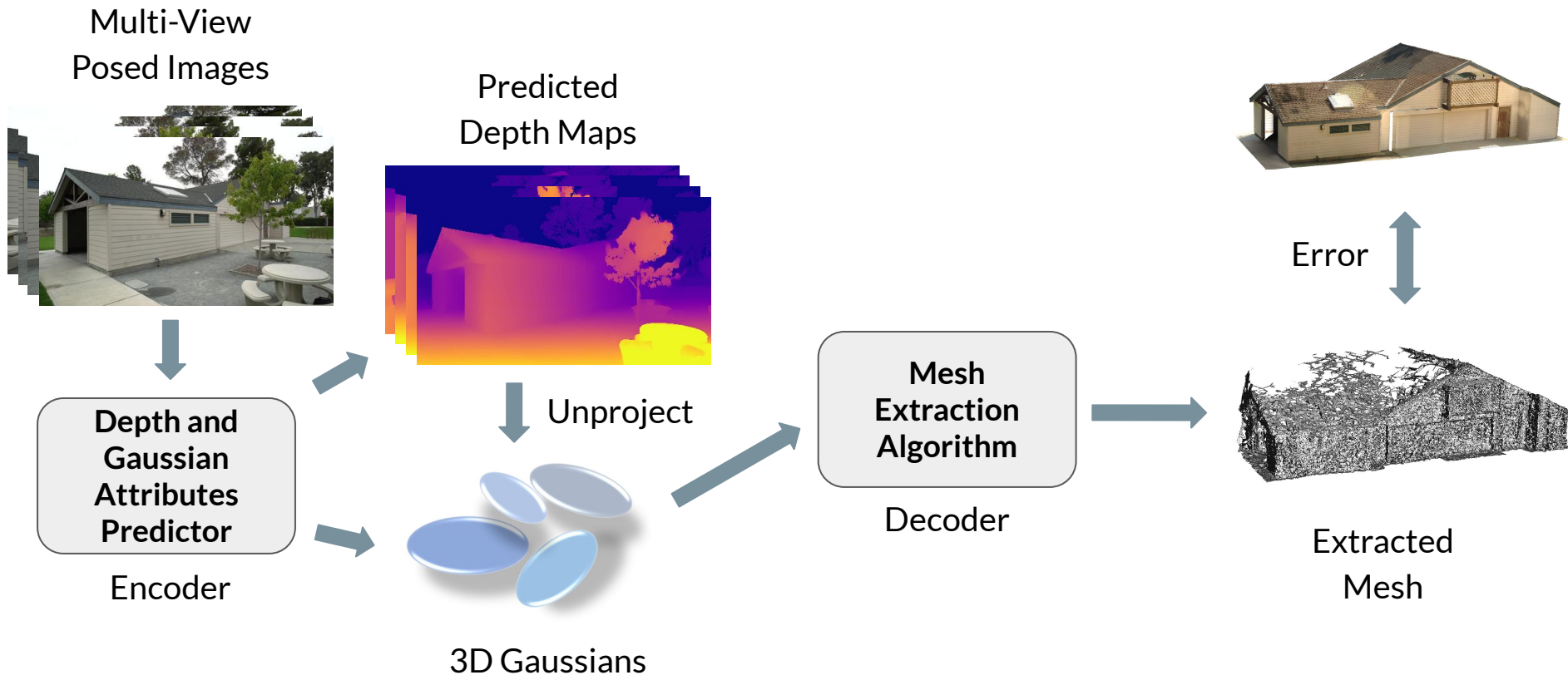
Main Hypothesis: Feed-Forward Gaussian Splatting models can achieve accurate 3D representations with photometric and self-supervised geometric losses

- 20% come from mixed datasets: indoor or outdoor
- GT depth comes from SfM methods thus incomplete

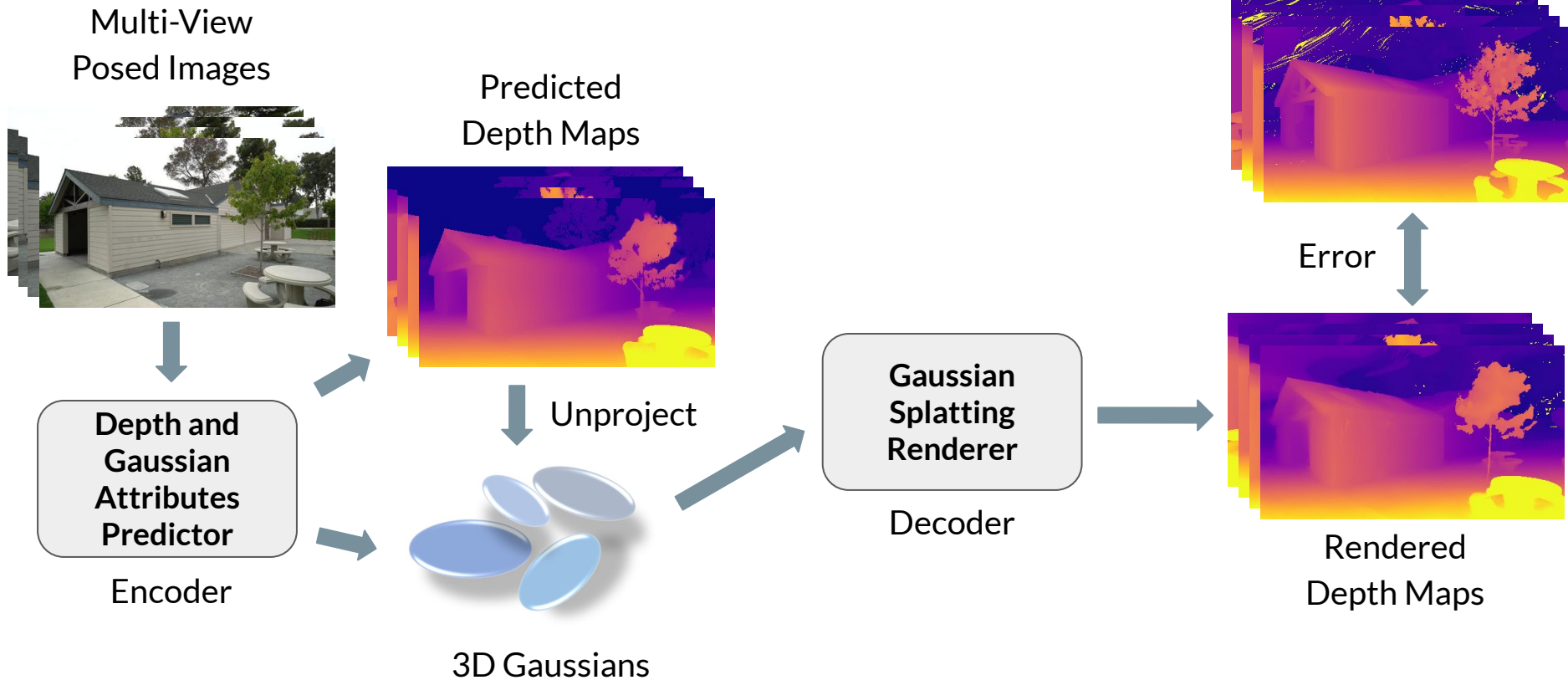
Framework: Training with NVS loss



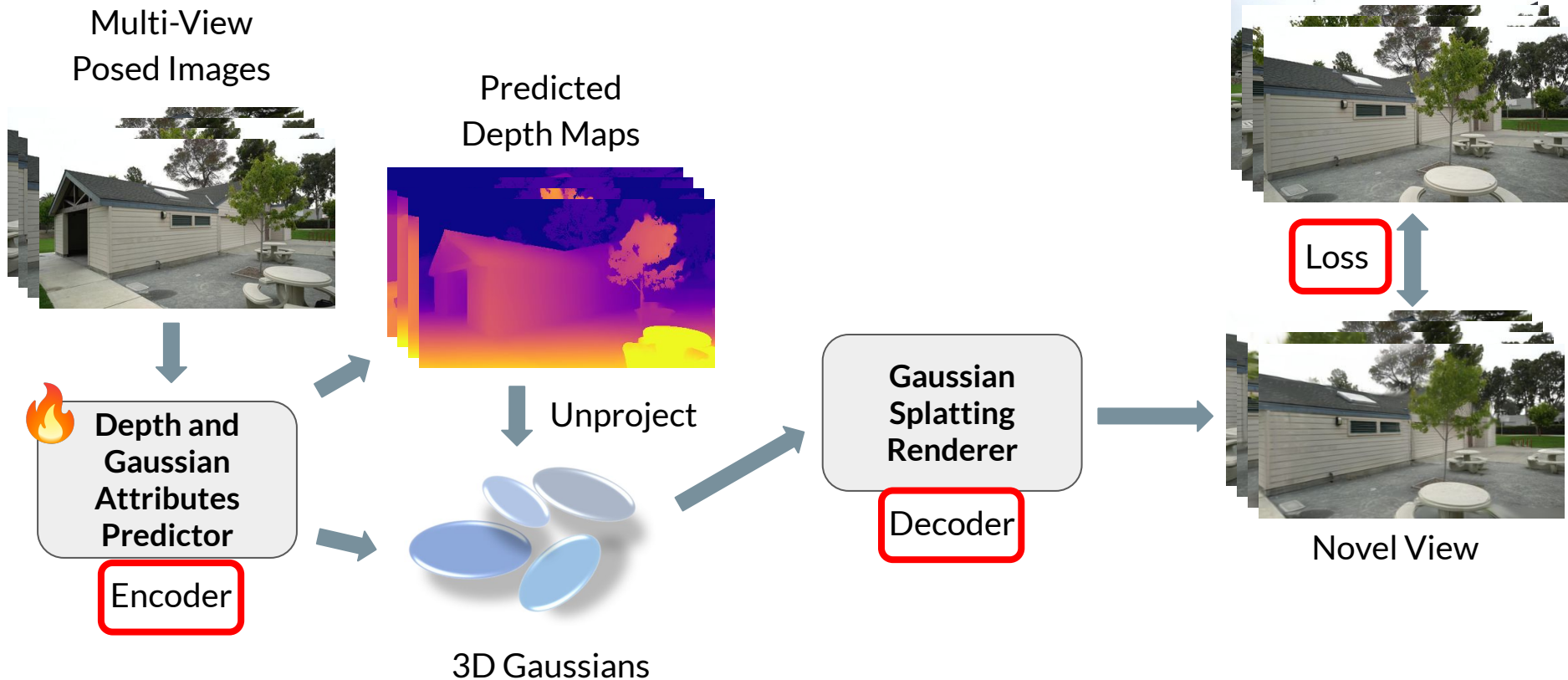
Framework: Mesh Extraction



Framework: Depth Estimation



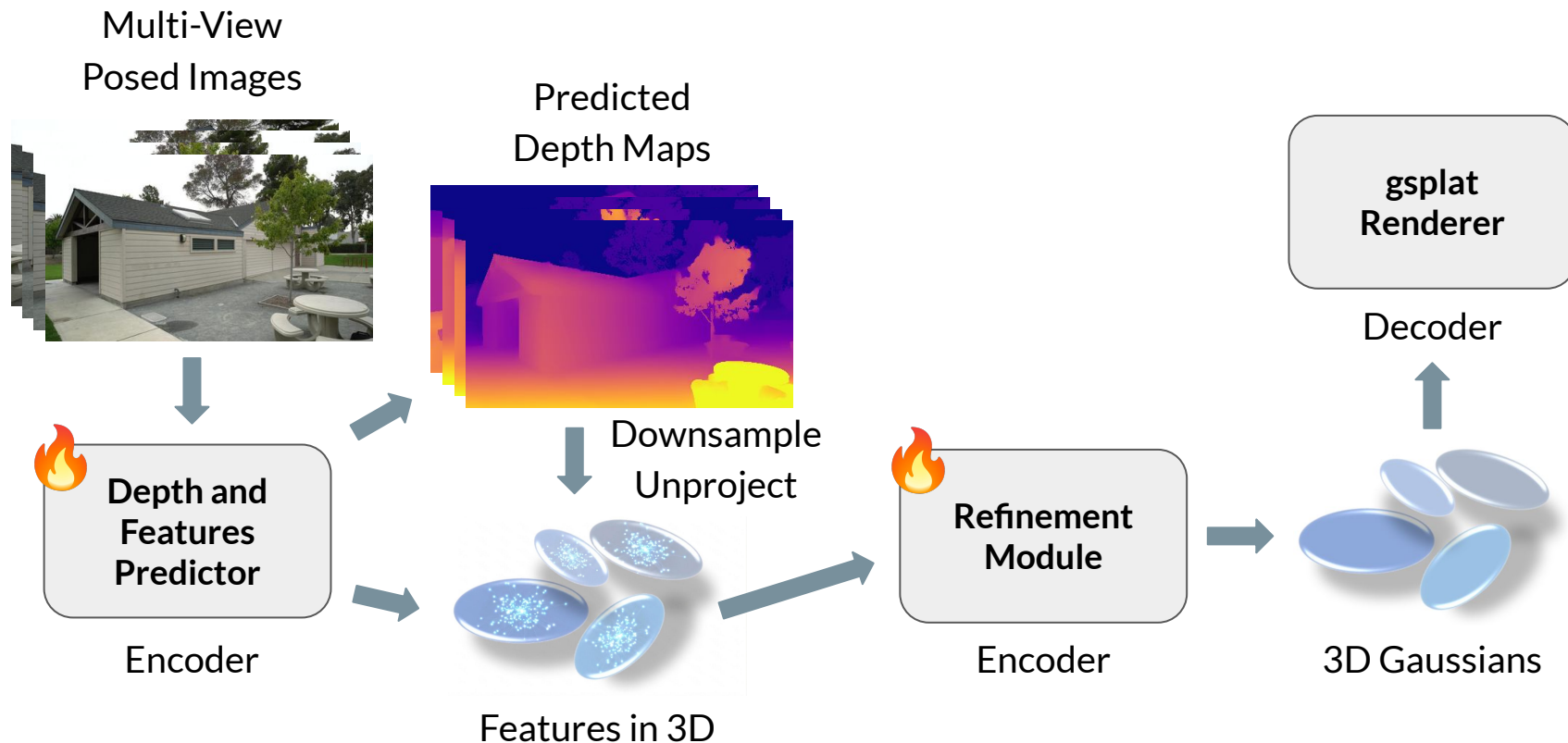
Framework: Training with NVS loss



2

Related Work

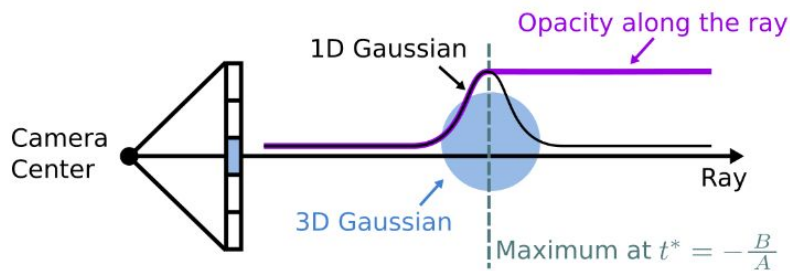
ReSplat: Learning Recurrent Gaussian Splatting



Gaussian Opacity Fields

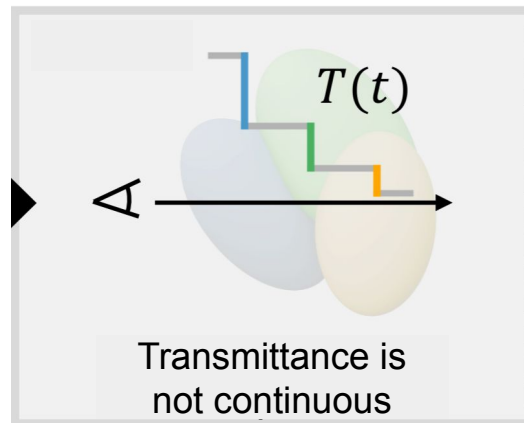
Contributions

- Color, depth, and normal renderings
- Defines opacity at every 3D position enabling unbounded mesh extraction



Shortcomings

- Depth rendering is not precise as it does not narrow down on 0.5-transmittance



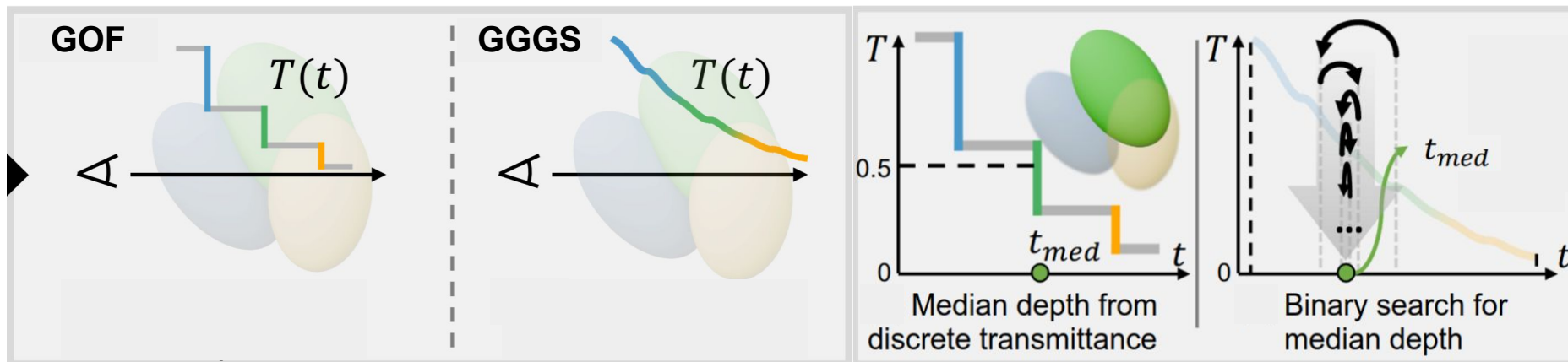
Geometry-Grounded Gaussian Splatting

Contributions

- Produces precise and consistent depth renderings

Shortcomings

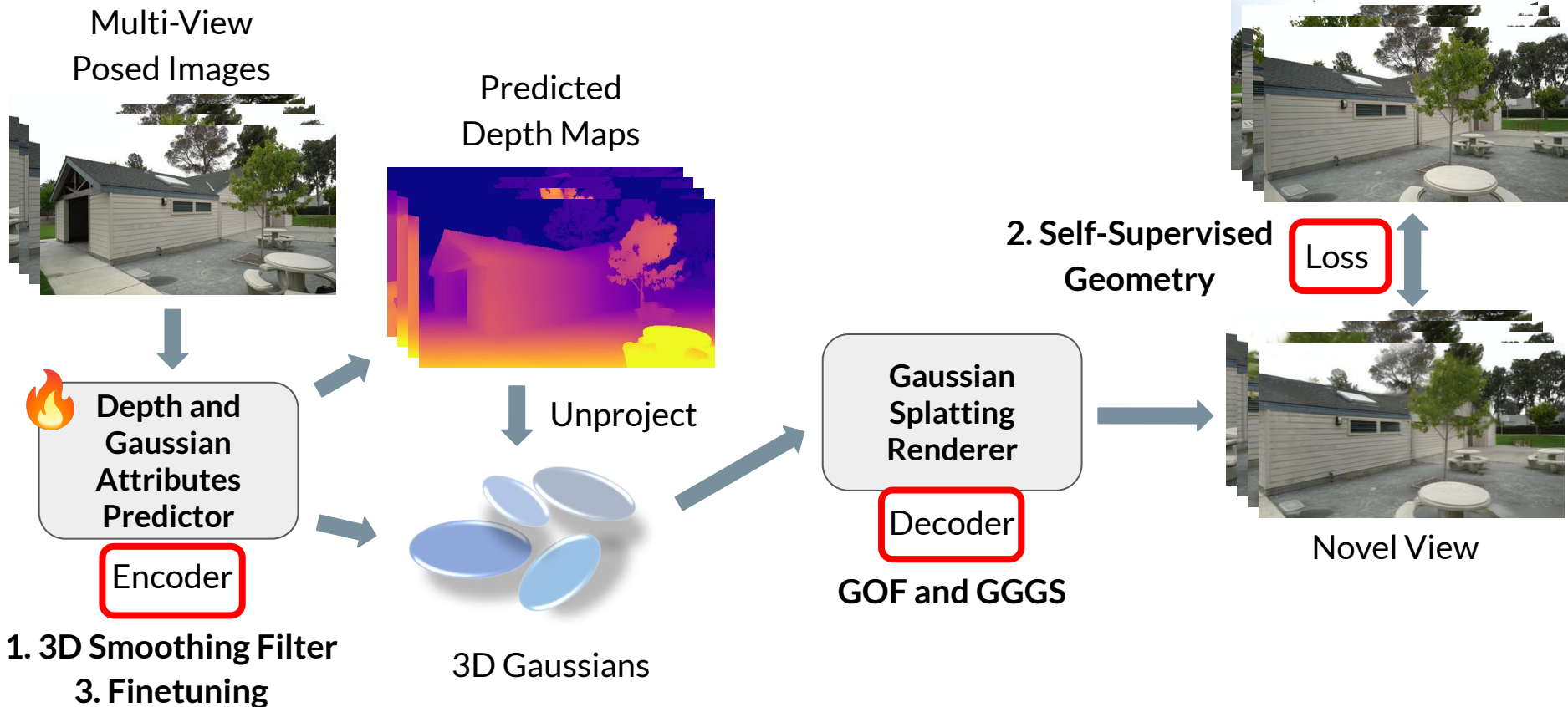
- Depth rendering process differs from color and normal rendering



3

Method

Framework: Additions



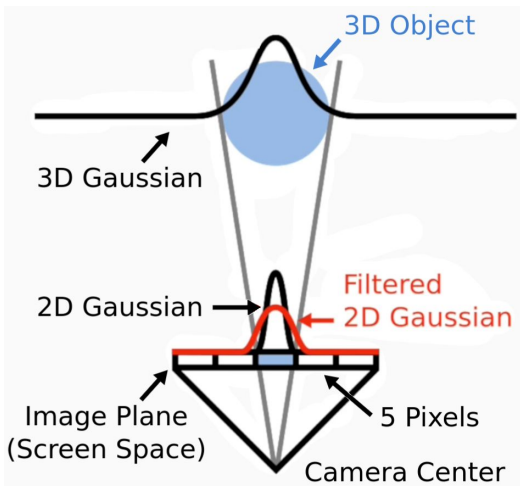
1. 3D Smoothing Filter



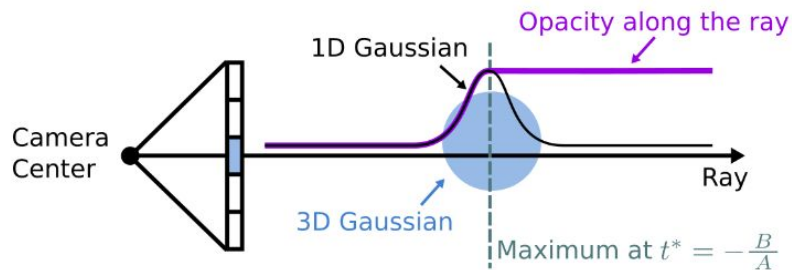
gsplat

GOF

1. 3D Smoothing Filter



gsplat



GOF

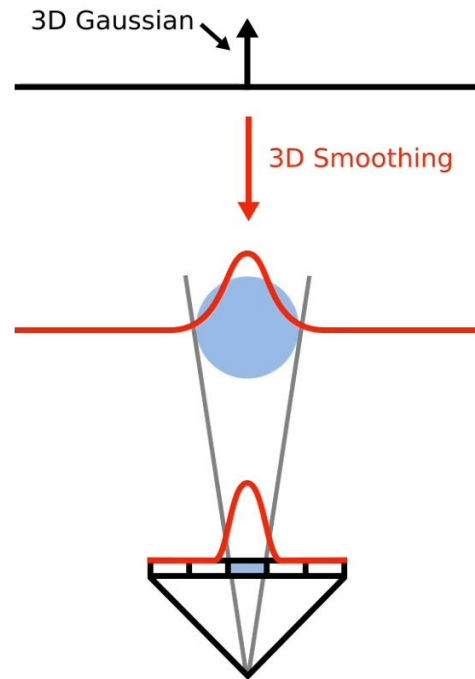
1. 3D Smoothing Filter

MipSplatting on Feed-Forward Gaussian Splatting

- **Goal:** Eliminate high-frequency artifacts thus stabilizing training when using GOF
- **Issue:** 3D Gaussian frequency exceeds the camera's sampling rate
- **Solution:** Apply a 3D smoothing filter to cap the Gaussian's frequency below the maximal sampling limit

constrain
frequency to $\nu_k = \frac{f_n}{d_{k,n}}$

achieved by $\Sigma_k = \Sigma_{k,\text{pred}} + \frac{s}{\nu_k} \cdot \mathbf{I}$



2. Self-Supervised Geometry Losses

Normal consistency loss from 2DGS

- Ensures gaussians are locally aligned with surfaces
- Encourages alignment between gaussian normals and the gradient of the depth map

$$\mathcal{L}_n = \sum_i \omega_i (1 - \mathbf{n}_i^T \mathbf{N})$$

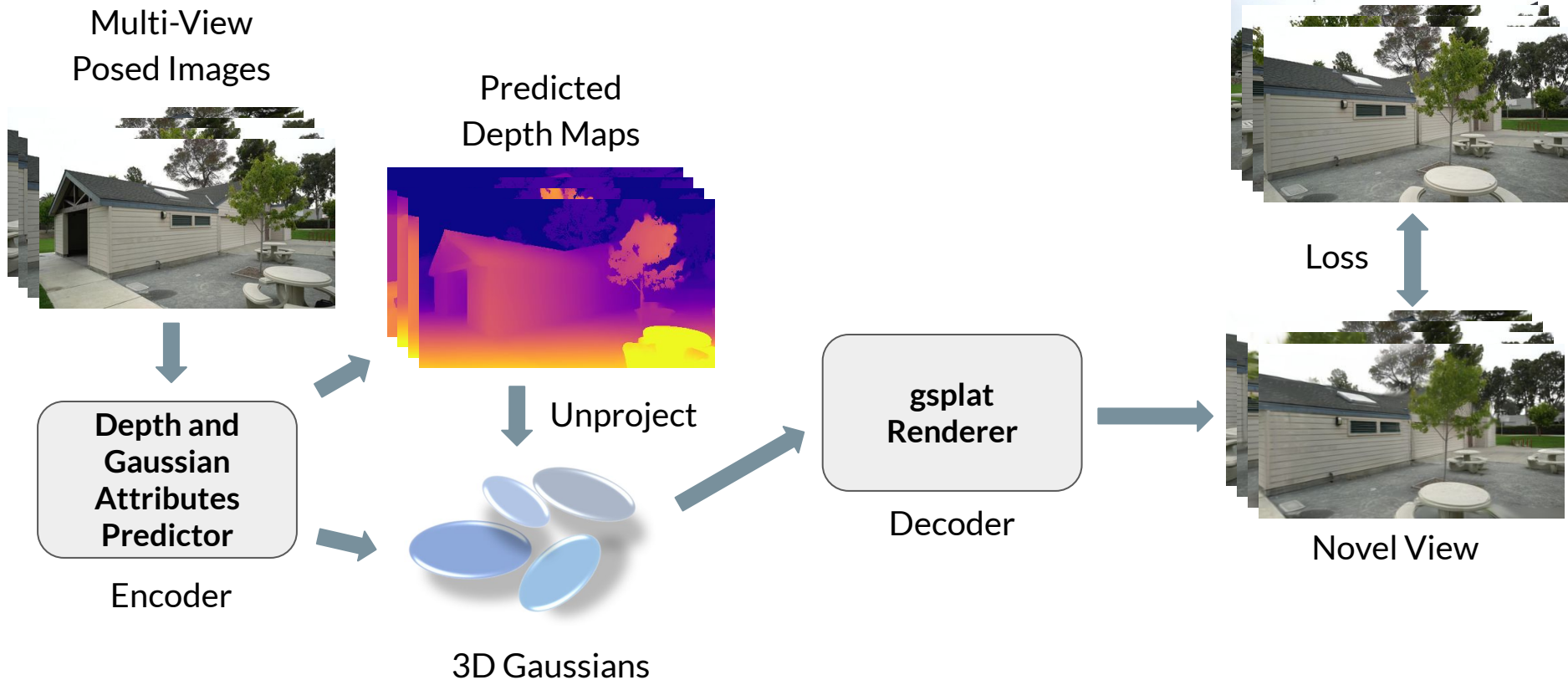
$$\mathbf{N}(x, y) = \frac{\nabla_x \mathbf{p}_s \times \nabla_y \mathbf{p}_s}{|\nabla_x \mathbf{p}_s \times \nabla_y \mathbf{p}_s|}$$

Other losses are unstable and redundant

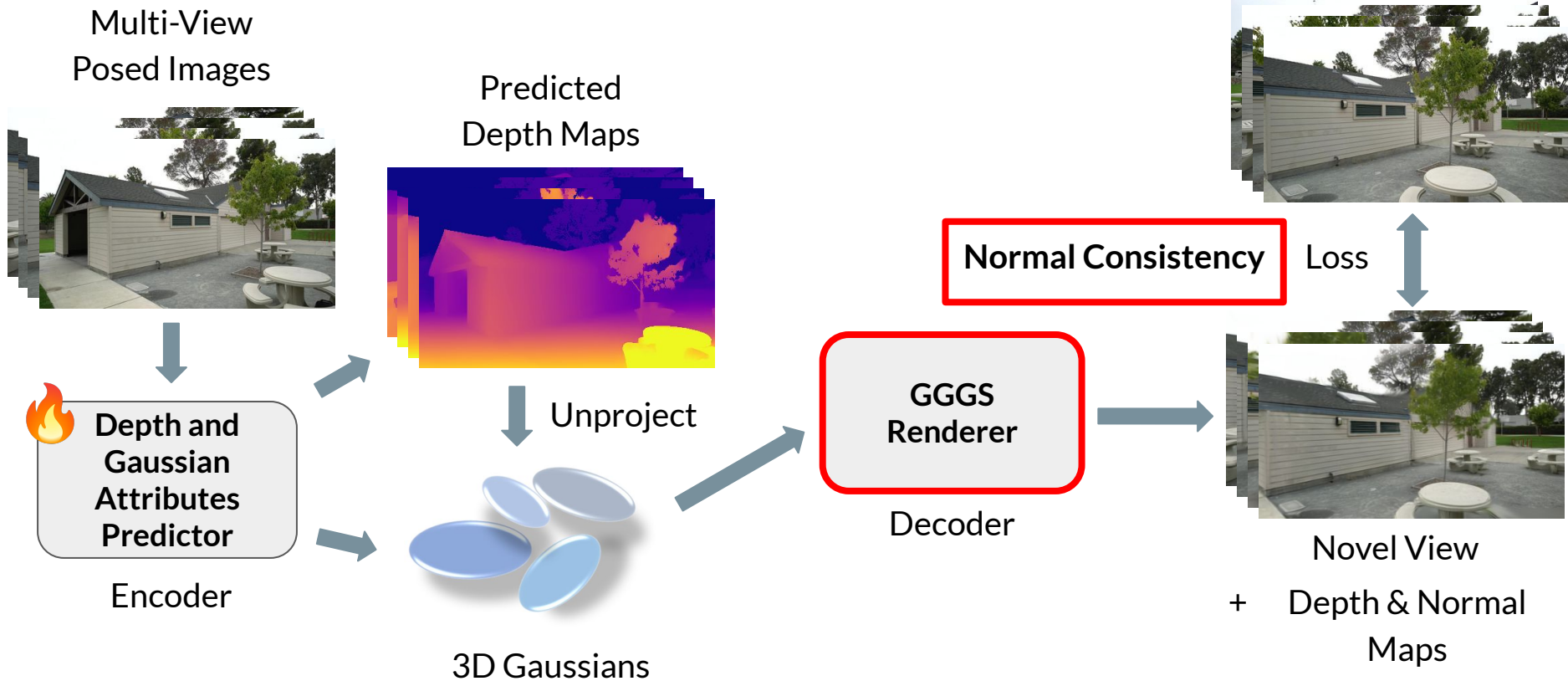
- Image Warp
- Edge-Aware Depth Smoothness

GGGS Opt.	w/o $\mathcal{L}_{iw}$	w/o $\mathcal{L}_n$	Full
F-Score $\uparrow$	0.60	0.57	0.60

3. ReSplat Encoder Fine-Tuning



3. ReSplat Encoder Fine-Tuning



3. ReSplat Encoder Fine-Tuning



3. ReSplat Encoder Fine-Tuning

Forward pass: Color rendering in GGGS and gsplat

- Project 3D Gaussians to 2D view-space Gaussians
- Filter 2D covariances to avoid stability issues

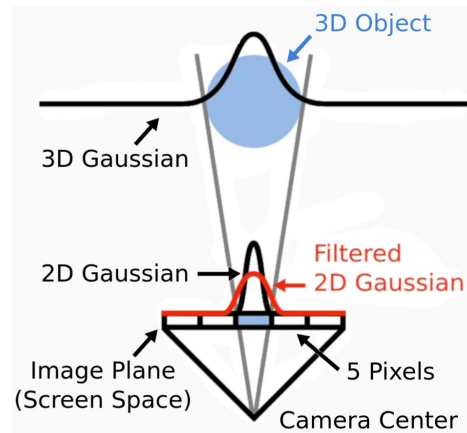
Backward pass: 2D Gaussian Filtering

GGGS

- Computes derivatives through intermediate variables
- Stops gradient flow when dividing by near-zero values

gsplat

- Simplifies chain rule math beforehand
- Avoids divisions which could lead to undefined or explosive gradients



3. ReSplat Encoder Fine-Tuning



4

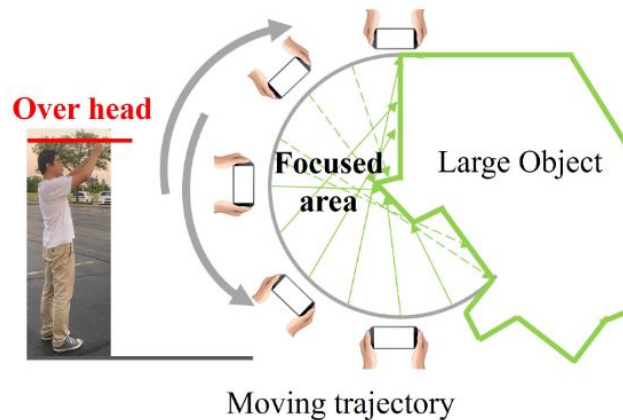
Evaluation

Datasets: Testing

- **Tanks and Temples:** Barn, Caterpillar, Courthouse, Ignatius, Meetingroom, Truck
- **MipNeRF360:** bicycle, flowers, garden, stump, treehill
- **DL3DV's** test set: 10 outdoor scenes

Context and target images

We select images which resemble the standardized camera trajectories of DL3DV



Methods

- **Encoder Depth Estimation:** Trained on RE10K and DL3DV with photometric loss and fine-tuned on indoor and synthetic datasets with GT depth supervision
- **Encoder**
 - Trained with gsplat as decoder and photometric losses
 - Trained from scratch with GOF and normal consistency loss
 - Fine-tuned to GOF decoder and normal consistency loss
 - Fine-tuned to GGGS decoder and normal consistency loss
- **GGGS per-scene optimization for 4k steps and regularization from 1k steps**
 - With MV losses: Including image-warp like loss
 - Without MV losses: L1, SSIM, and normal consistency loss
- **SurfSplat**
- **Depth Anything 3 Giant**

Metrics

Mesh Reconstruction

Tanks and Temples provides GT point clouds and evaluation code. We report:

- F-Score: Percentage of points in reconstruction close enough to the ground truth and viceversa

Depth Estimation

‘GT’ depth maps are rendered from the per-scene GGGS-optimized gaussian scene. We report:

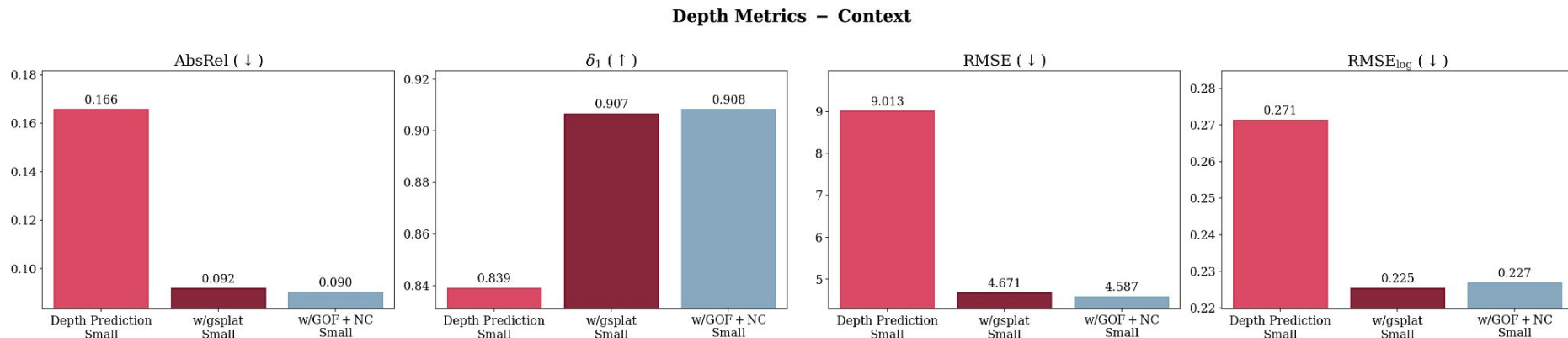
- Mean Absolute Relative L1 error
- RMSE and log RMSE

- Threshold Accuracy, δ_1

$$\max \left(\frac{d_{\text{gt}}}{d_{\text{rend}}}, \frac{d_{\text{rend}}}{d_{\text{gt}}} \right) < 1.25$$

Finding 1:

Encoder Trained with Photometric Loss Excels at Depth Accuracy



- Training with photometric and normal consistency losses using GOF decoder does not significantly improve depth accuracy

Finding 1:

Encoder Trained with Photometric Loss Excels at Depth Accuracy

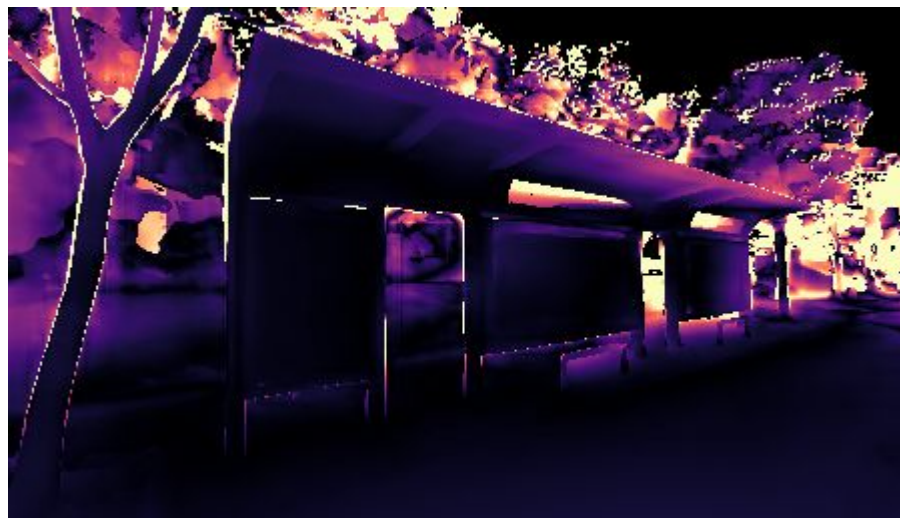
Input Images



Finding 1:

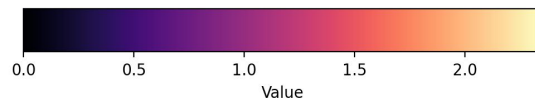
Encoder Trained with Photometric Loss Excels at Depth Accuracy

Absolute Depth Difference, non-metric



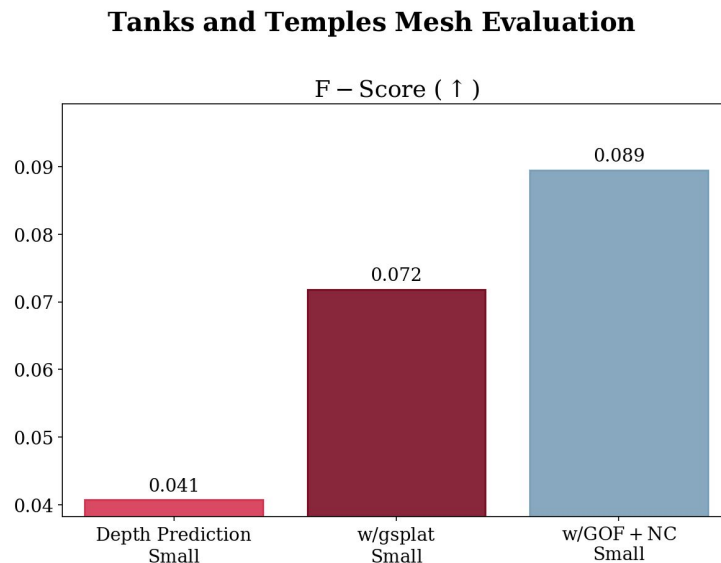
Scene AbsRel:

Depth Predictor
0.1829



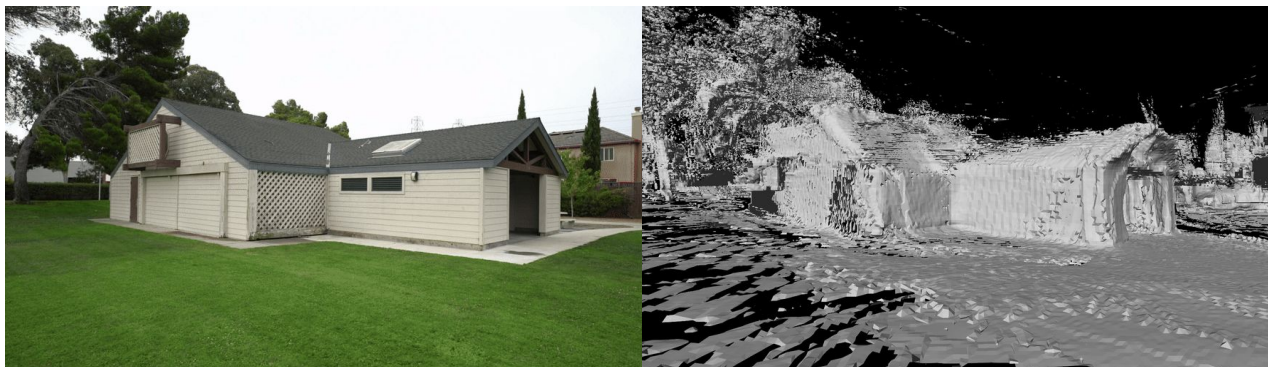
Encoder w/gsplat
0.0612

Finding 2: Photometric and Normal Consistency Loss Enable Accurate Mesh Reconstruction



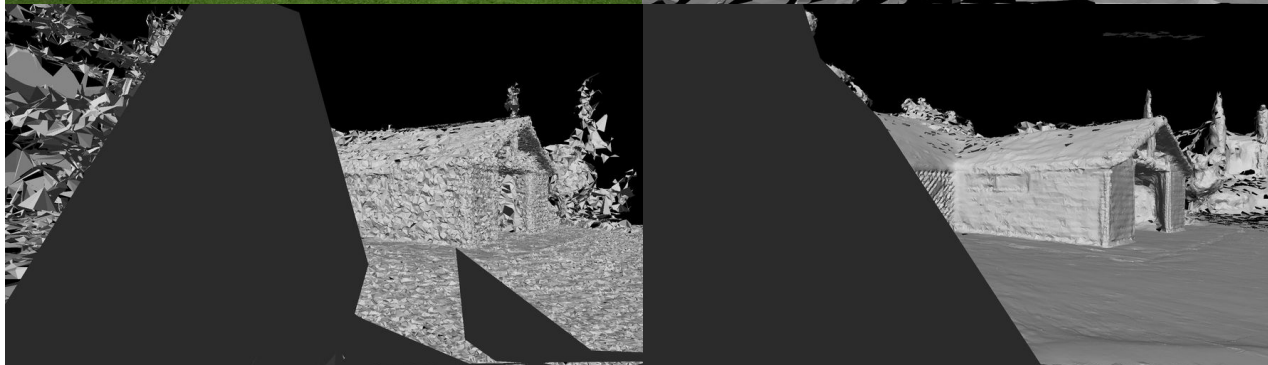
Finding 2: Photometric and Normal Consistency Loss Enable Accurate Mesh Reconstruction

Color



Depth
Predictor
F-Score:
0.0335

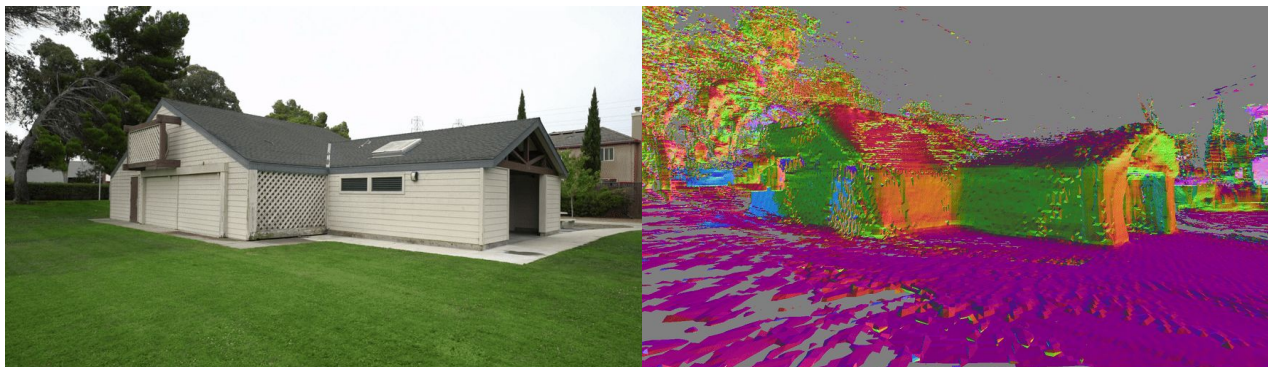
w/gsplat
Small
F-Score:
0.0682



w/GOF
Small
F-Score:
0.0738

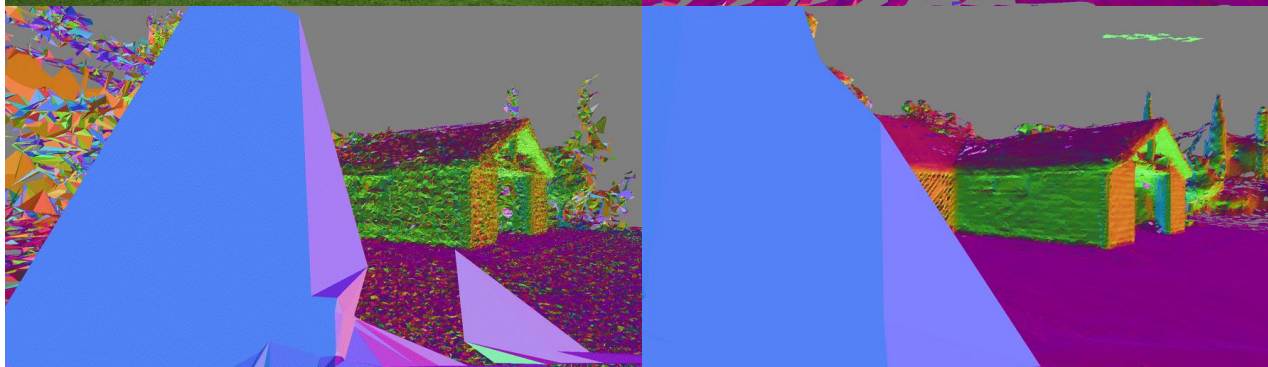
Finding 2: Photometric and Normal Consistency Loss Enable Accurate Mesh Reconstruction

Color



Depth
Predictor
F-Score:
0.033

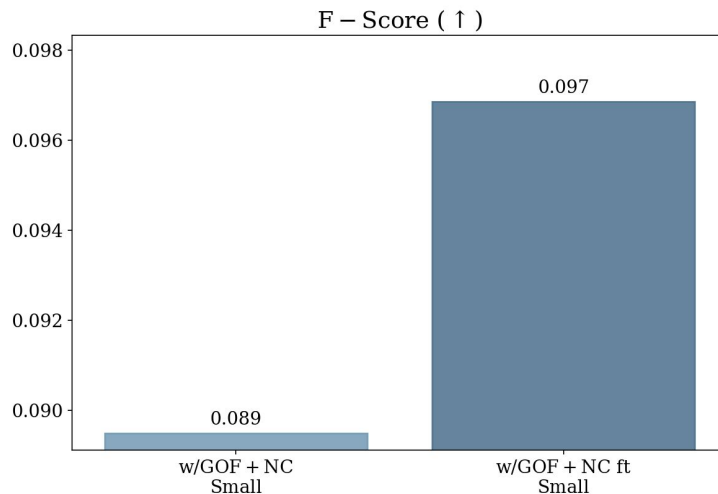
w/gsplat
Small
F-Score:
0.068



w/GOF
Small
F-Score:
0.073

Finding 3: Fine-Tuning on Normal Consistency Loss Outperforms Training from Scratch

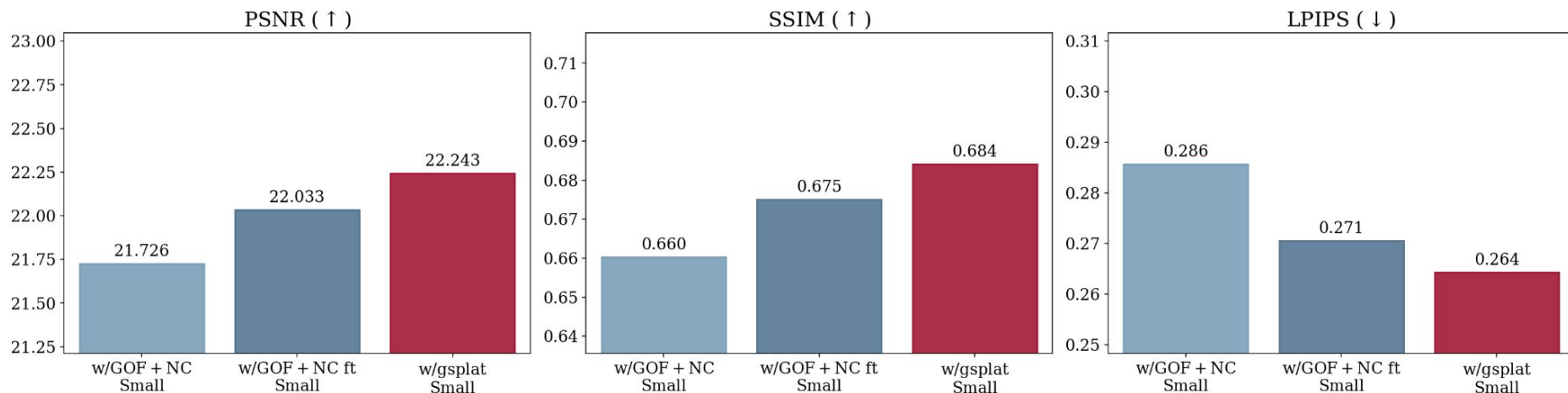
Tanks and Temples Mesh Evaluation



Finding 3:

Fine-Tuning on Normal Consistency Loss Outperforms Training from Scratch

NVS Metrics – Target



Finding 3: Fine-Tuning on Normal Consistency Loss Outperforms Training from Scratch

GT



w/GOF+NC
ft Small

Target views
PSNR: 20.30



w/GOF+NC
Small

Target views
PSNR: 20.00

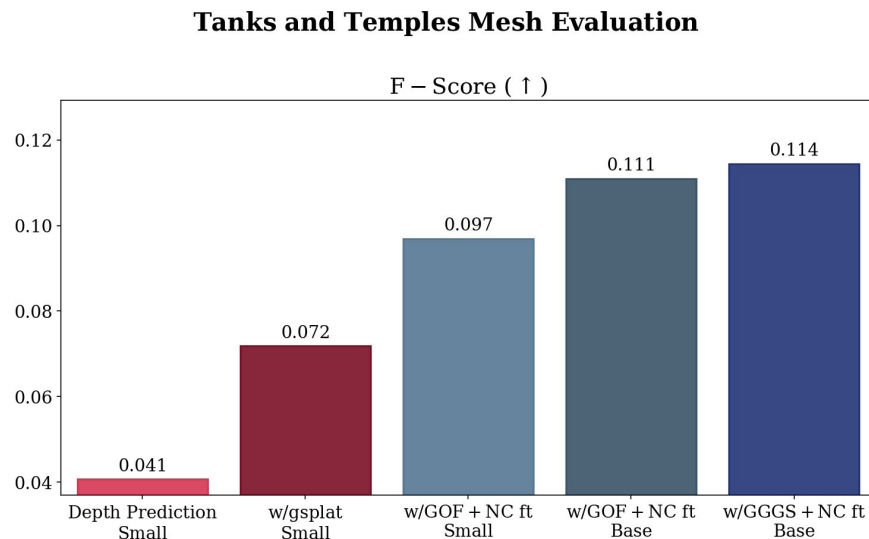


w/gsplat
Small

Target views
PSNR: 20.49



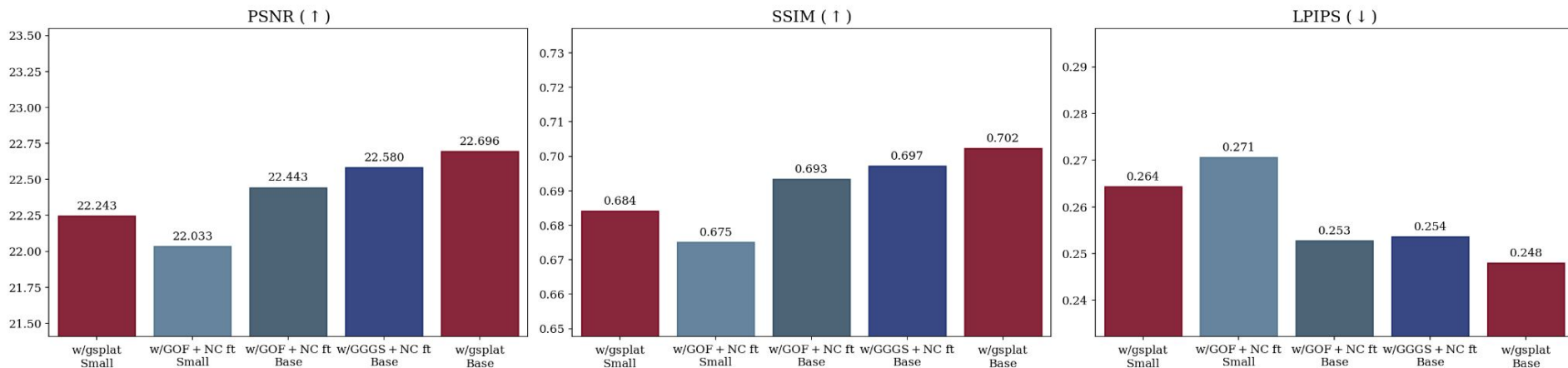
Finding 4: Fine-Tuned Base Encoder with GGGS and Normal Consistency Loss Performs the Best



Finding 4:

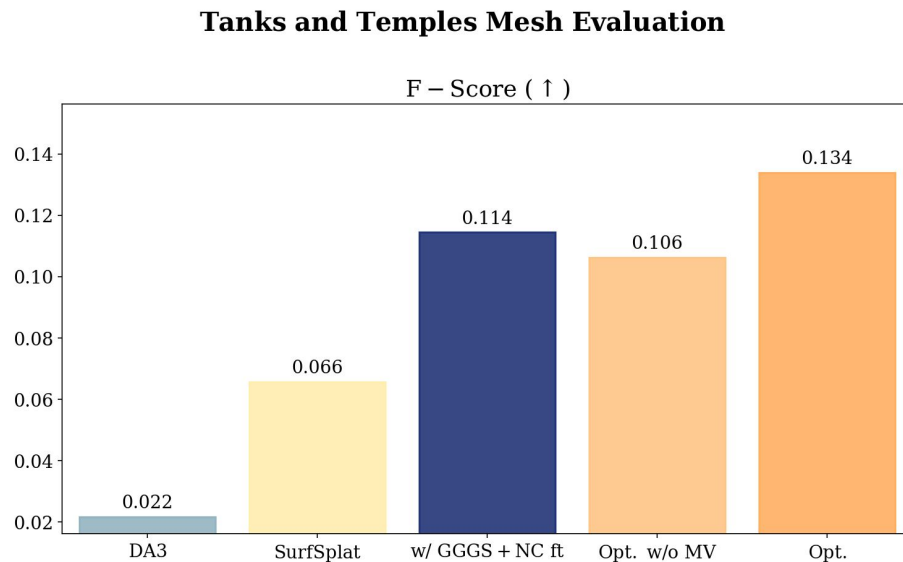
Fine-Tuned Base Encoder with GGGS and Normal Consistency Loss Performs the Best

NVS Metrics – Target



Finding 5:

Fine-Tuned Base Encoder Outperforms DA3, SurfSplat, and Performs on-par with per-scene Optimization

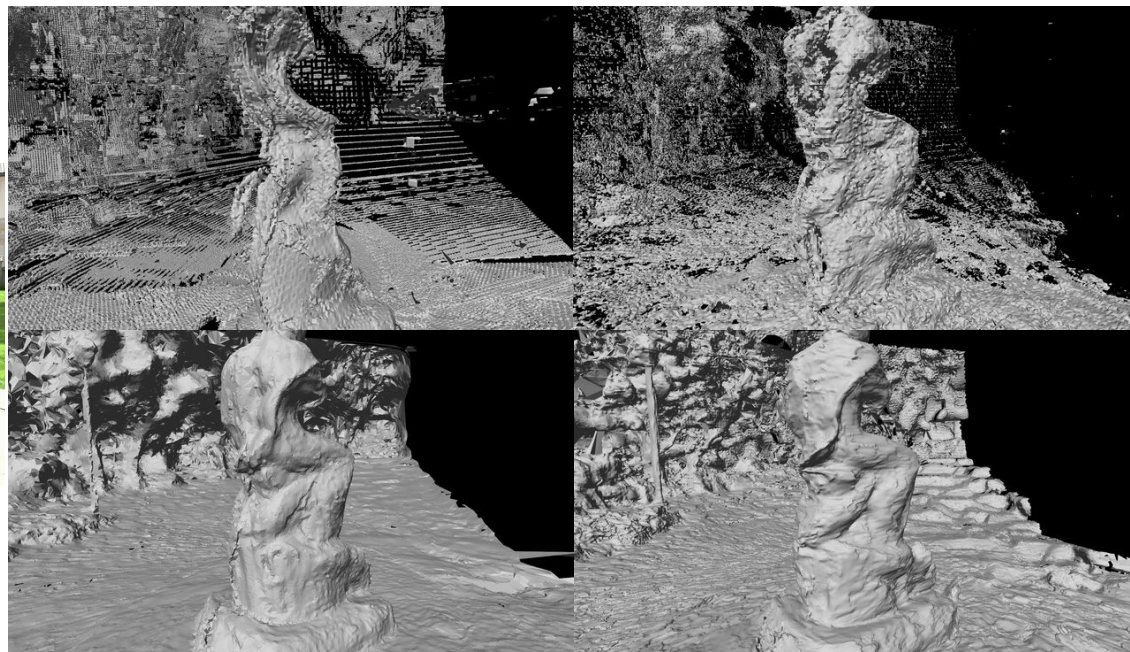


Finding 5: Fine-Tuned ReSplat Encoder Outperforms DA3, SurfSplat, and Performs on-par with per-scene Optimization

DA3. F-Score: 0.03

SurfSplat. F-Score: 0.12

Color



w/GGGS+NC ft. F-Score: 0.17

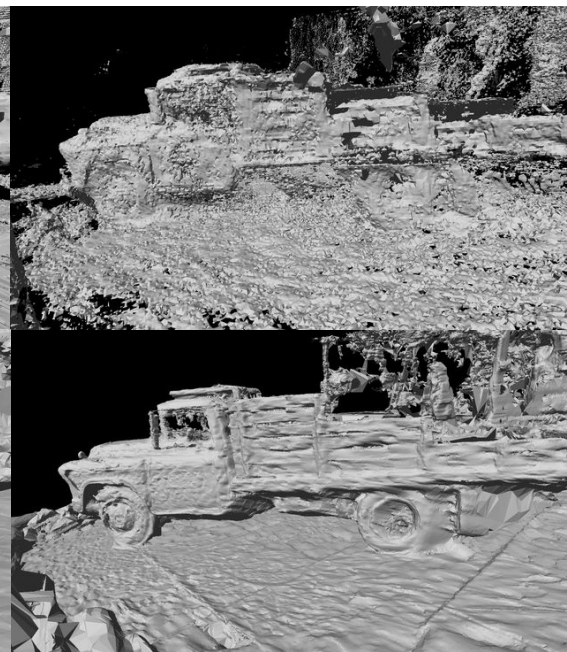
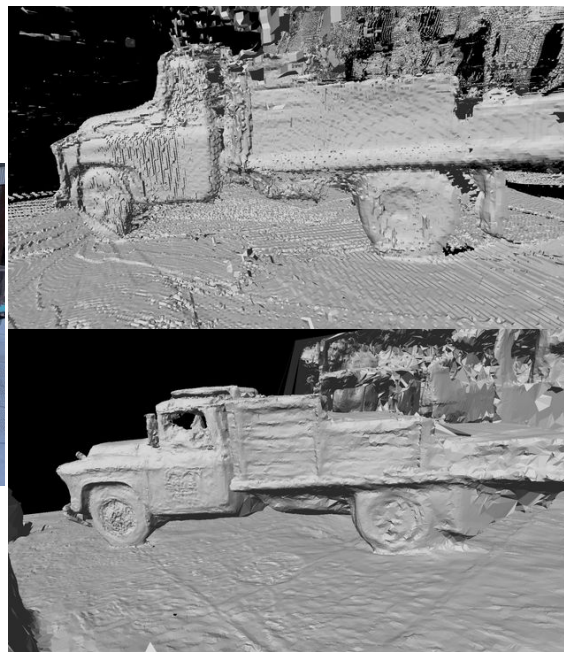
GGGS Opt. F-Score: 0.26

Finding 5: Fine-Tuned ReSplat Encoder Outperforms DA3, SurfSplat, and Performs on-par with per-scene Optimization

DA3. F-Score: 0.05

SurfSplat. F-Score: 0.14

Color

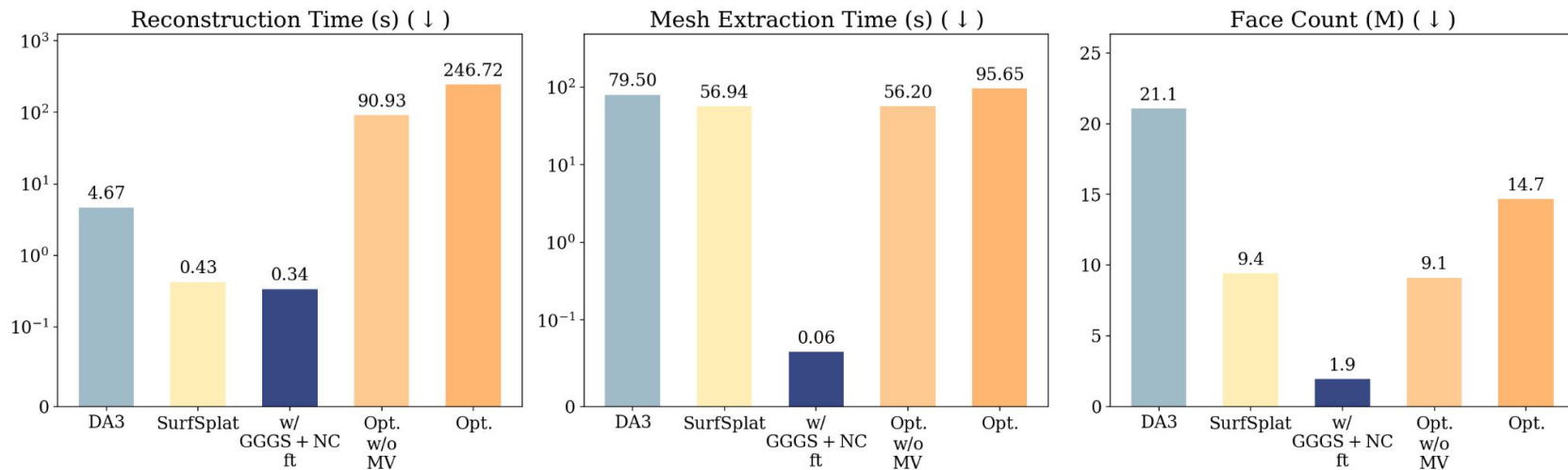


w/GGGS+NC ft. F-Score: 0.18

GGGS Opt. F-Score: 0.19

Finding 5: Fine-Tuned ReSplat Encoder Outperforms DA3, SurfSplat, and Performs on-par with per-scene Optimization

Metrics on Tanks and Temples



5

Conclusion

Conclusion

1. Feed-Forward Gaussian Splatting models **can** achieve accurate **depth estimation** with photometric losses
2. Normal consistency loss does not improve depth estimation significantly but does improve **mesh reconstruction accuracy**
3. Fine-Tuning Feed-Forward Gaussian Splatting models is not straightforward and depends on the choice of **primitives** and **renderers**
4. Feed-Forward Gaussian Splatting models **achieve state-of-the-art performance in depth estimation and mesh reconstruction tasks**

Future Work:

- Compact 3D representations circumvent depth predictor module
- Primitives other than gaussians: triangles or meshes
- New depth priors: Depth Anything V3



Fin

Questions ?

Appendix

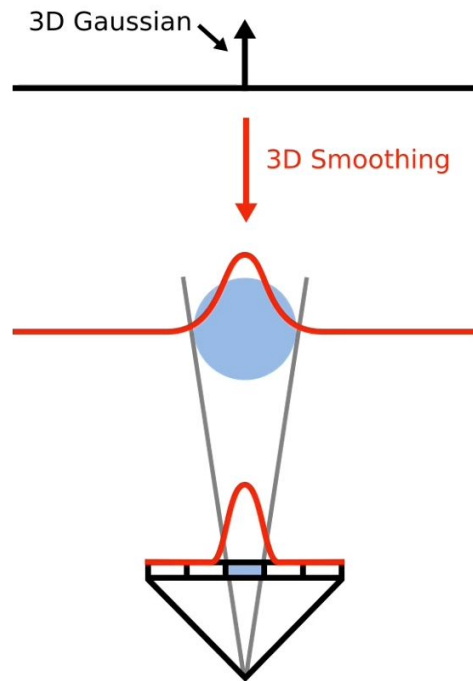
1. 3D Smoothing Filter

MipSplatting on 3D Gaussian Splatting

- **Goal:** Eliminate high-frequency artifacts
- **Issue:** 3D Gaussian frequency exceeds the camera's sampling rate
- **Solution:** Apply a 3D smoothing filter to cap the Gaussian's frequency below the maximal sampling limit

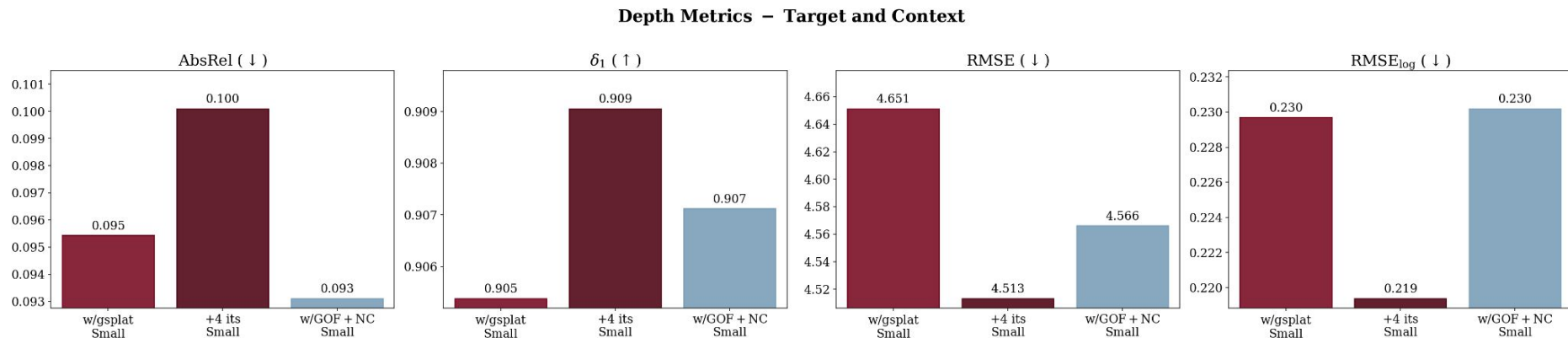
constrain
frequency to

$$\hat{\nu}_k = \max \left(\left\{ \mathbb{1}_n(\mathbf{p}_k) \cdot \frac{f_n}{d_n} \right\}_{n=1}^N \right)$$



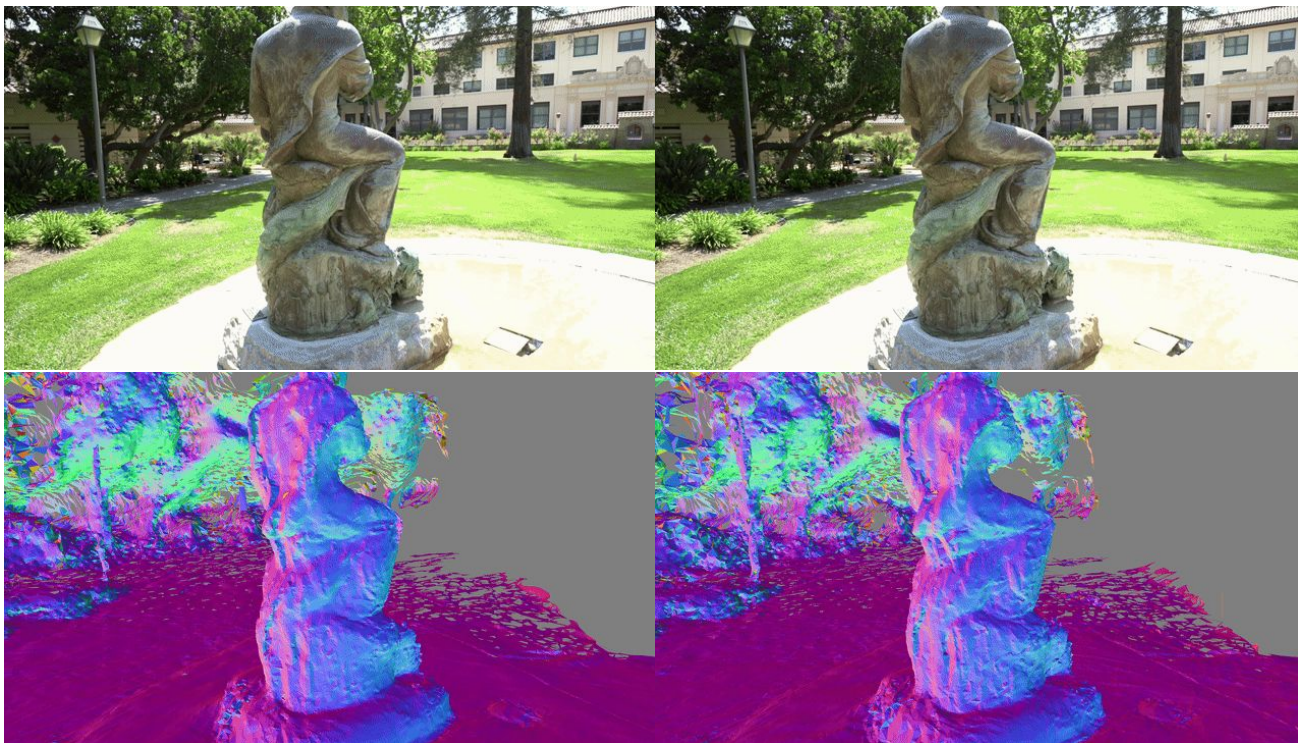
Finding 1:

Encoder Trained with Photometric Loss Excels at Depth Accuracy



- Refinement module also does not significantly improve depth accuracy

Finding 3: Fine-Tuning on Normal Consistency Loss Outperforms Training from Scratch

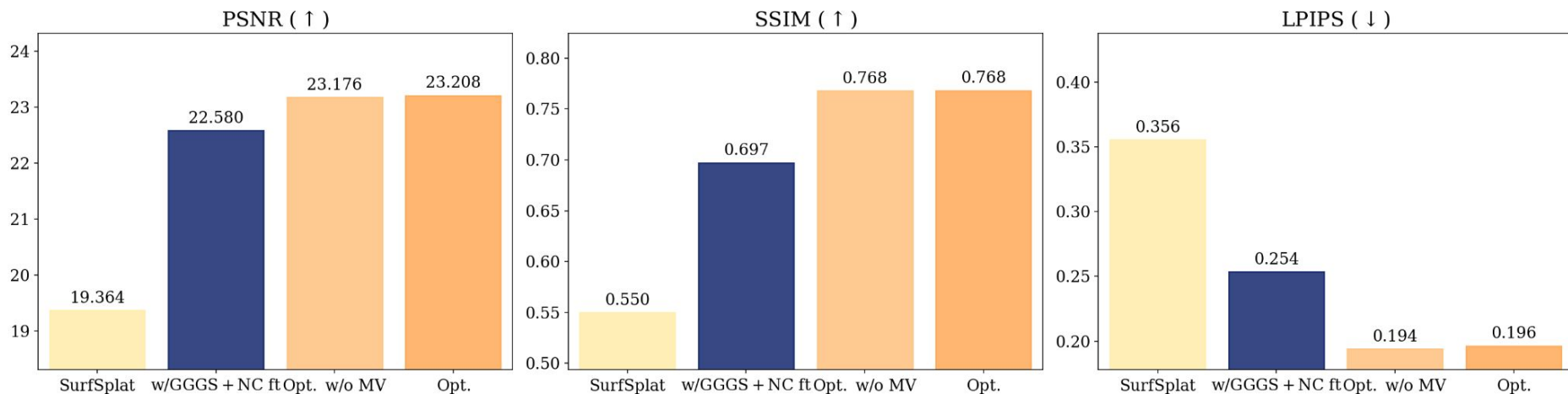


Finding 3: Fine-Tuning on Normal Consistency Loss Outperforms Training from Scratch



Finding 5: Fine-Tuned ReSplat Encoder Outperforms DA3, SurfSplat, and Performs on-par with per-scene Optimization

NVS Metrics – Target



Abstract

Existing feed-forward Gaussian Splatting models focus on visual quality, while the few approaches targeting geometric quality are limited to indoor scenes and image pairs. This thesis focuses on enabling feed-forward models to achieve geometrically accurate 3D representations of unbounded scenes from 8 or 16 sparse views at 256 by 448 resolution, evaluated via extracted meshes and rendered depth maps. Leveraging the recent ReSplat model, we make three major contributions towards this goal:

First, we stabilize training by applying a 3D filter to the predicted Gaussians, preventing them from exceeding the Nyquist frequency. Second, we explore the behavior of common self-supervised losses for geometric accuracy in feed-forward models. We find that edge-aware depth smoothness and image warp losses lead to unstable training. Instead, we use an edge-aware normal consistency loss.

Third, we successfully fine-tune a ReSplat encoder, trained with gsplat as its decoder, using the Gaussian Opacity Fields and the Geometry-Grounded Gaussian Splatting (GGGS) decoders alongside the edge-aware normal consistency loss. These fine-tuned encoders achieve similar geometric accuracy and higher visual quality compared to training from scratch.

We evaluate our GGGS fine-tuned encoder and observe strong performance across three key metrics. For mesh reconstruction on the Tanks and Temples dataset, our method improves precision by 73% over the photometric-only encoder, performs on par with per-scene optimization, and surpasses Depth Anything 3 by a factor of three. For depth estimation across MipNeRF360, Tanks and Temples, and DL3DV, our method reduces absolute relative distance by 7% against the photometric-only baseline, 31% against Depth Anything 3, and performs on par with per-scene optimization. These improvements come at a minimal cost to visual fidelity, maintaining a PSNR within 0.12dB of the photometric-only encoder.